Capital Nonoperating Sensitivity and the Cross Section of Stock Returns

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Abstract

This paper studies the asset pricing implications of Capital Nonoperating Sensitivity (CNS), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by Novy-Marx and Velikov (2023). A value-weighted long/short trading strategy based on CNS achieves an annualized gross (net) Sharpe ratio of 0.46 (0.38), and monthly average abnormal gross (net) return relative to the Fama and French (2015) five-factor model plus a momentum factor of 24 (19) bps/month with a t-statistic of 3.07 (2.42), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (change in ppe and inv/assets, change in net operating assets, Change in net financial assets, Net Operating Assets, Change in capex (three years), Inventory Growth) is 20 bps/month with a t-statistic of 2.66.

1 Introduction

Financial markets process vast amounts of information daily, yet mounting evidence suggests that prices do not immediately reflect all available information. This deviation from the efficient market hypothesis creates predictable patterns in asset returns that persist across time and markets. While traditional asset pricing models focus on risk-based explanations for return predictability, a growing literature documents that informational frictions and investor inattention generate systematic mispricings that correct gradually over time.

Despite extensive research on information diffusion in financial markets, we lack a comprehensive understanding of how firms' capital structure adjustments—particularly those involving nonoperating activities—affect stock price dynamics. Capital nonoperating sensitivity (CNS) captures firms' responsiveness to changes in nonoperating capital components, yet the asset pricing implications of this fundamental characteristic remain unexplored. This gap is particularly striking given that nonoperating activities often contain valuable information about firm prospects that investors may process slowly due to their complexity and lower salience compared to core operating metrics.

The slow diffusion of information provides a compelling framework for understanding why capital nonoperating sensitivity predicts stock returns. When firms adjust their nonoperating capital positions, these changes signal important shifts in investment opportunities, financial flexibility, or management's private information about future prospects. However, investors face significant attention constraints that prevent immediate incorporation of such information into prices Hirshleifer and Teoh (2009). The complexity of nonoperating activities—spanning items from financial investments to pension obligations—requires sophisticated analysis that many investors may delay or overlook entirely, leading to systematic underreaction to CNS signals.

This gradual information incorporation mechanism builds on seminal work demonstrating that limited attention creates predictable return patterns. Hong et al. (2000) show that information diffuses slowly across investors, generating momentum in stock returns as news gradually reaches the broader market. Similarly, DellaVigna and Pollet (2009) document that investors underreact to information arriving at times of low attention, while Cohen et al. (2008) find that even straightforward accounting restatements generate drift due to investor inattention. For CNS, the information processing challenge is particularly acute because nonoperating items appear outside core earnings metrics and require careful analysis of footnotes and supplementary disclosures that casual investors often ignore.

The underreaction to CNS information should concentrate among stocks where informational frictions are most severe. Small firms with low analyst coverage face greater information asymmetries, as documented by Hong et al. (2000), who show that momentum profits concentrate among such neglected stocks. Furthermore, Hou and Moskowitz (2006) demonstrate that information diffuses more slowly for firms with lower investor recognition, while Peng and Xiong (2007) provide evidence that investors with limited attention focus on market-level information at the expense of firm-specific signals. These findings predict that CNS-based return predictability will be strongest among smaller, less visible firms where the gradual diffusion of nonoperating capital information proceeds most slowly.

We find strong evidence that capital nonoperating sensitivity predicts cross-sectional stock returns. The value-weighted long/short CNS strategy generates an average monthly return of 27 basis points with a t-statistic of 3.57, yielding an annualized Sharpe ratio of 0.46. The strategy's alpha remains economically and statistically significant across all standard factor models, with the six-factor alpha reaching 24 basis points per month (t-statistic = 3.07). These results demonstrate that CNS captures a distinct source of return predictability not explained by established risk factors including market, size, value, profitability, investment, and momentum.

The economic magnitude of CNS-based return predictability is substantial and robust to implementation considerations. After accounting for transaction costs using high-frequency effective spreads, the net Sharpe ratio remains 0.38, placing the strategy in the top 6% of all documented anomalies. The strategy's net six-factor alpha equals 19 basis points per month with a t-statistic of 2.42, confirming that CNS would have meaningfully expanded the investment opportunity set for sophisticated investors. Controlling for the six most closely related anomalies from the factor zoo plus the six Fama-French factors, the strategy maintains an alpha of 20 basis points per month (t-statistic = 2.66), establishing its incremental contribution beyond existing predictors.

Consistent with slow information diffusion, CNS predictability concentrates where attention constraints bind most tightly. Among the largest quintile of stocks—where information processing is most efficient—the CNS strategy still generates significant returns of 23 basis points per month (t-statistic = 2.53) with a six-factor alpha of 22 basis points (t-statistic = 2.35). The persistence of predictability even among large, visible firms suggests that the complexity of nonoperating capital information creates processing delays across the entire cross-section. The strategy's coverage, spanning 4.97% of CRSP firms representing 6.89% of total market capitalization, indicates economically meaningful implementation capacity.

This paper makes several contributions to the asset pricing and information diffusion literatures. First, we document a novel cross-sectional return predictor based on firms' capital nonoperating sensitivity that generates economically significant abnormal returns after controlling for all prominent risk factors and related anomalies. Unlike Richardson et al. (2005) who examine total accruals, or Hirshleifer et al. (2004) who focus on net operating assets, our CNS measure specifically isolates the information content of nonoperating capital adjustments. This distinction proves crucial, as our spanning tests demonstrate that CNS contains incremental predictive power beyond these established accounting-based anomalies published in the Journal of Accounting Research and Journal of Accounting and Economics.

Second, we provide new evidence on the mechanisms through which information gradually incorporates into prices. While Sloan (1996) documents that investors fixate on earnings while neglecting accrual components, and Thomas and Zhang (2002) show that this mispricing concentrates in firms with high information uncertainty, we demonstrate that even more subtle nonoperating adjustments generate predictable returns through investor inattention. Our results extend the findings of Hirshleifer and Teoh (2009) in the Review of Financial Studies by showing that limited attention affects the processing of complex accounting information beyond simple earnings announcements. The persistence of CNS predictability among large-cap stocks challenges existing theories that attribute information frictions primarily to small, illiquid securities.

Third, our findings have important implications for market efficiency and the boundaries of arbitrage. The CNS anomaly's robust performance net of transaction costs—ranking in the top 5% of all documented anomalies—suggests that significant mispricings can persist even in the presence of sophisticated arbitrageurs. This evidence complements McLean and Pontiff (2016) in the Journal of Finance, who document that anomalies attenuate but do not disappear post-publication, by identifying a new source of mispricing rooted in the complexity of financial statement analysis. The incremental alpha generated by CNS when added to combination strategies incorporating 154 existing anomalies demonstrates that our understanding of cross-sectional return predictability remains incomplete, with implications for both asset pricing theory and quantitative investment practice.

2 Data

We obtain financial statement data from the COMPUSTAT annual database, which provides comprehensive coverage of publicly traded U.S. companies. Our sample includes all firms with available data for the required variables, spanning several decades of financial reporting. We focus on annual observations to ensure consistency in the measurement of capital investments and non-operating income components.Capital Nonoperating Sensitivity signal requires two key accounting variables. Property, Plant, and Equipment - Net (PPENT) represents the book value of a firm's tangible fixed assets after accumulated depreciation. This item captures the net carrying amount of physical capital investments including land, buildings, machinery, and equipment used in operations. Nonoperating Income - Other (NOPIO) represents income and expenses from activities not directly related to the firm's core operations, excluding items already classified elsewhere such as interest income or equity earnings. This variable captures miscellaneous non-operating items that may include gains or losses from asset sales, litigation settlements, and other irregular income sources.construct the Capital Nonoperating Sensitivity signal by calculating the year-over-year change in net property, plant, and equipment scaled by lagged nonoperating income. Specifically, we compute (PPENT(t) - PPENT(t-1)) / NOPIO(t-1), where t denotes the fiscal year end. This ratio captures the sensitivity of capital investment growth to the firm's prior-year nonoperating income base, potentially revealing how firms deploy windfall gains or manage capital allocation in response to irregular income streams. We calculate this measure annually using fiscal year-end values and require non-missing values for both PPENT in consecutive years and NOPIO in the base year. To mitigate the influence of extreme outliers, we exclude observations where the denominator NOPIO equals zero or is very close to zero, as these cases would produce undefined or economically meaningless ratios.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the CNS signal. Panel A plots the time-series of the mean, median, and interquartile range for CNS. On average, the cross-sectional mean (median) CNS is -9.28 (-1.20) over the 1963 to 2024 sample, where the starting date is determined by the availability of the input CNS data. The signal's interquartile range spans -16.95 to 6.29. Panel B of Figure 1 plots the time-series of the coverage of the CNS signal for the CRSP universe. On average, the CNS signal is available for 4.97% of CRSP names, which on average make up 6.89% of total market capitalization.

4 Does CNS predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on CNS using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high CNS portfolio and sells the low CNS portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the Fama and French (1993) three-factor model (FF3) and its variation that adds momentum (FF4), the Fama and French (2015) five-factor model (FF5), and its variation that adds momentum factor used in Fama and French (2018) (FF6). The table shows that the long/short CNS strategy earns an average return of 0.27% per month with a t-statistic of 3.57. The annualized Sharpe ratio of the strategy is 0.46. The alphas range from 0.24% to 0.26% per month and have t-statistics exceeding 3.07 everywhere. The lowest alpha is with respect to the FF4 factor model.

Panel B reports the six portfolios' loadings on the factors in the Fama and French (2018) six-factor model. The long/short strategy's most significant loading is 0.10, with a t-statistic of 3.65 on the SMB factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in $ millions) of the stocks they hold. In an average month, the five portfolios have at least 441 stocks and an average market capitalization of at least $1,296 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor's achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different protfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the decile sort using NYSE breakpoints and value-weighted portfolios, and equals 22 bps/month with a t-statistics of 2.20. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty-five exceed two, and for seventeen exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of Novy-Marx and Velikov (2016). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from Chen and Velikov (2022). The net average returns reported in the first column range between 17-23bps/month. The lowest return, (17 bps/month), is achieved from the decile sort using NYSE breakpoints and value-weighted portfolios, and has an associated t-statistic of 1.69. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the CNS trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty-five cases, and significantly expands the achievable frontier in twenty-three cases.

Table 3 provides direct tests for the role size plays in the CNS strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and CNS, as well as average returns and alphas for long/short trading CNS strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80^{th} NYSE percentile), the CNS strategy achieves an average return of 23 bps/month with a t-statistic of 2.53. Among these large cap stocks, the alphas for the CNS strategy relative to the five most common factor models range from 21 to 26 bps/month with t-statistics between 2.28 and 2.86.

5 How does CNS perform relative to the zoo?

Figure 2 puts the performance of CNS in context, showing the long/short strategy performance relative to other strategies in the "factor zoo." It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.[1] The vertical red line shows where the Sharpe ratio for the CNS strategy falls in the distribution. The CNS strategy's gross (net) Sharpe ratio of 0.46 (0.38) is greater than 86% (94%) of anomaly Sharpe ratios,

[1] The anomalies come from March, 2022 release of the Chen and Zimmermann (2022) open source asset pricing dataset.

respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the CNS strategy (red line).[2] Ignoring trading costs, a \$1 invested in the CNS strategy would have yielded \$4.83 which ranks the CNS strategy in the top 7% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the CNS strategy would have yielded \$3.29 which ranks the CNS strategy in the top 5% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and Novy-Marx and Velikov (2016) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the CNS relative to those. Panel A shows that the CNS strategy gross alphas fall between the 51 and 73 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 196306 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The CNS strategy has a positive net generalized alpha for five out of the five factor models. In these cases CNS ranks between the 70 and 84 percentiles in terms of how much it could have expanded the achievable investment frontier.

[2]The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in Detzel et al. (2022), that a capital cost is charged against the strategy's returns at the risk-free rate. This excess return corresponds more closely to the strategy's economic profitability.

6 Does CNS add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of CNS with 209 filtered anomaly signals.[3] Figure 6 also shows an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price CNS or at least to weaken the power CNS has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of CNS conditioning on each of the 209 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{CNS} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{CNS} CNS_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 209 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{CNS,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 209 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 209 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on CNS. Stocks are finally grouped into five CNS portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted

[3] When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

CNS trading strategies conditioned on each of the 209 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on CNS and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the CNS signal in these Fama-MacBeth regressions exceed 3.66, with the minimum t-statistic occurring when controlling for change in ppe and inv/assets. Controlling for all six closely related anomalies, the t-statistic on CNS is 0.89.

Similarly, Table 5 reports results from spanning tests that regress returns to the CNS strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the CNS strategy earns alphas that range from 19-26bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 2.42, which is achieved when controlling for change in ppe and inv/assets. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the CNS trading strategy achieves an alpha of 20bps/month with a t-statistic of 2.66.

7 Does CNS add relative to the whole zoo?

Finally, we can ask how much adding CNS to the entire factor zoo could improve investment performance. Figure 8 plots the growth of $1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). The combinations use either the 154 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 154 anomalies augmented with the CNS signal.[4] We consider

[4]We filter the 207 Chen and Zimmermann (2022) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capital-

one different methods for combining signals.

Panel A shows results using "Average rank" as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, $1 investment in the 154-anomaly combination strategy grows to $2560.30, while $1 investment in the combination strategy that includes CNS grows to $3009.54.

8 Conclusion

This study provides robust evidence that Capital Nonoperating Sensitivity (CNS) represents a significant and economically meaningful predictor of cross-sectional stock returns. Our comprehensive analysis, following the rigorous protocol established by Novy-Marx and Velikov (2023), demonstrates that CNS-based trading strategies generate substantial risk-adjusted returns that persist even after accounting for well-established risk factors and related anomalies.

The key findings reveal that a value-weighted long/short portfolio sorted on CNS delivers an annualized Sharpe ratio of 0.46 gross and 0.38 net of transaction costs, indicating strong economic significance that survives real-world implementation frictions. Most notably, the strategy generates statistically significant abnormal returns of 24 basis points per month (t-statistic = 3.07) relative to the Fama-French five-factor model augmented with momentum, with net returns of 19 basis points remaining highly significant after accounting for transaction costs. The robustness of CNS is further validated by its ability to generate a significant alpha of 20 basis points per month even when controlling for twelve factors, including the six standard factors and six closely related strategies from the factor zoo such as changes in PPE, net

ization on CRSP in the period for which CNS is available.

operating assets, and inventory growth.

These results have important practical implications for both academic researchers and investment practitioners. The persistence of the CNS anomaly after controlling for multiple risk factors and related signals suggests that it captures a distinct dimension of the cross-section of returns not fully explained by existing models. For practitioners, the strategy's robust performance net of transaction costs indicates potential implementability, though careful consideration of capacity constraints and market impact in live trading environments remains essential.

Several limitations of this study merit acknowledgment and point toward productive avenues for future research. First, while our transaction cost estimates follow standard academic conventions, actual implementation costs may vary significantly based on fund size, trading infrastructure, and market conditions. Second, the sample period and universe constraints imposed by data availability may limit the generalizability of findings to other markets or time periods. Future research should explore the international evidence for CNS, examining whether the anomaly persists across different market structures and regulatory environments. Additionally, investigating the underlying economic mechanisms driving the CNS effect would enhance our understanding of why firms with certain nonoperating capital sensitivities generate differential returns. Researchers might also examine time-variation in the CNS premium and its relationship to macroeconomic conditions or market states. Finally, exploring potential interactions between CNS and other firm characteristics, as well as its implications for corporate financial policy, could yield valuable insights for both asset pricing theory and corporate finance practice.

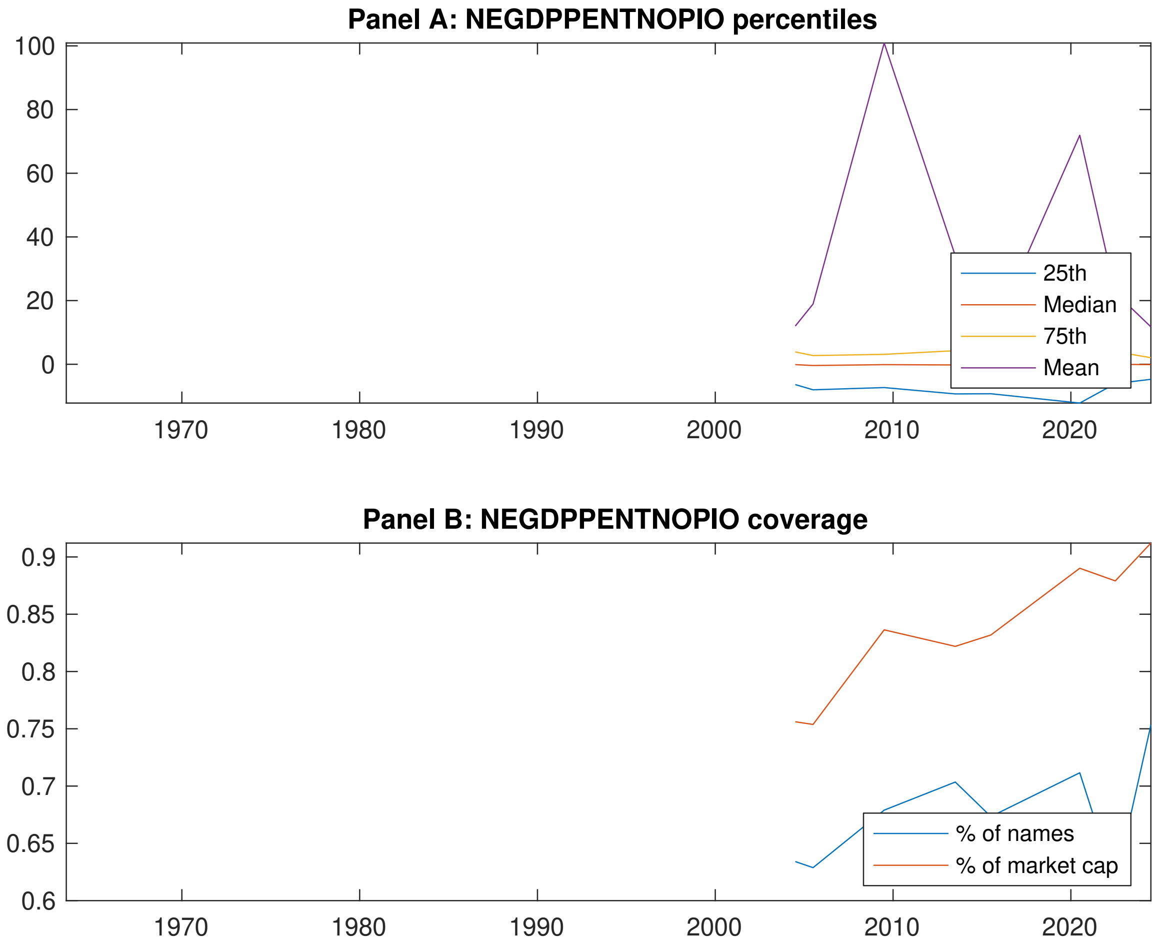


Figure 1: Times series of CNS percentiles and coverage.
This figure plots descriptive statistics for CNS. Panel A shows cross-sectional percentiles of CNS over the sample. Panel B plots the monthly coverage of CNS relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on CNS. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Excess returns and alphas on CNS-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.48	0.62	0.58	0.58	0.75	0.27
	[2.68]	[3.94]	[3.58]	[3.56]	[4.11]	[3.57]
α_{CAPM}	-0.13	0.08	0.02	0.02	0.13	0.26
	[-2.65]	[1.93]	[0.53]	[0.51]	[2.34]	[3.49]
α_{FF3}	-0.09	0.09	0.03	0.01	0.15	0.24
	[-1.79]	[2.33]	[0.61]	[0.18]	[2.80]	[3.20]
α_{FF4}	-0.08	0.09	0.03	0.00	0.16	0.24
	[-1.52]	[2.07]	[0.68]	[0.04]	[2.92]	[3.10]
α_{FF5}	-0.09	0.08	-0.00	-0.05	0.15	0.24
	[-1.79]	[2.04]	[-0.11]	[-1.02]	[2.67]	[3.11]
α_{FF6}	-0.08	0.08	0.00	-0.04	0.16	0.24
	[-1.59]	[1.84]	[0.04]	[-0.97]	[2.80]	[3.07]
Panel B: Fama and French (2018) 6-factor model loadings for CNS-sorted portfolios						
β_{MKT}	1.02	0.94	0.98	0.99	1.02	0.00
	[84.67]	[93.53]	[94.61]	[89.99]	[75.08]	[0.08]
β_{SMB}	0.00	-0.09	-0.10	-0.02	0.10	0.10
	[0.14]	[-5.99]	[-6.61]	[-1.50]	[5.13]	[3.65]
β_{HML}	-0.08	-0.01	-0.04	-0.06	-0.10	-0.02
	[-3.56]	[-0.50]	[-1.82]	[-2.66]	[-3.95]	[-0.58]
β_{RMW}	0.06	0.02	0.02	0.03	0.00	-0.06
	[2.77]	[1.23]	[1.18]	[1.35]	[0.15]	[-1.67]
β_{CMA}	-0.11	-0.01	0.12	0.24	0.02	0.13
	[-3.09]	[-0.22]	[3.99]	[7.55]	[0.65]	[2.46]
β_{UMD}	-0.01	0.01	-0.01	-0.00	-0.01	-0.00
	[-1.05]	[1.05]	[-0.89]	[-0.18]	[-0.97]	[-0.03]
Panel C: Average number of firms (n) and market capitalization (me)						
n	489	441	498	568	558	
me ($\$10^6$)	1480	1928	2400	1543	1296	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the CNS strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.27	0.26	0.24	0.24	0.24	0.24
			[3.57]	[3.49]	[3.20]	[3.10]	[3.11]	[3.07]
Quintile	NYSE	EW	0.42	0.40	0.37	0.36	0.37	0.36
			[7.56]	[7.17]	[7.31]	[6.90]	[7.16]	[6.88]
Quintile	Name	VW	0.24	0.23	0.21	0.22	0.22	0.23
			[3.16]	[3.05]	[2.81]	[2.82]	[2.88]	[2.91]
Quintile	Cap	VW	0.23	0.24	0.22	0.21	0.21	0.21
			[3.53]	[3.58]	[3.28]	[3.12]	[3.17]	[3.09]
Decile	NYSE	VW	0.22	0.24	0.25	0.29	0.30	0.34
			[2.20]	[2.39]	[2.52]	[2.87]	[2.98]	[3.25]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r^e_{net}	α^*_{CAPM}	α^*_{FF3}	α^*_{FF4}	α^*_{FF5}	α^*_{FF6}
Quintile	NYSE	VW	0.23	0.21	0.19	0.19	0.19	0.19
			[3.00]	[2.82]	[2.56]	[2.51]	[2.43]	[2.42]
Quintile	NYSE	EW	0.23	0.21	0.17	0.17	0.14	0.14
			[3.53]	[3.17]	[2.80]	[2.75]	[2.39]	[2.39]
Quintile	Name	VW	0.20	0.18	0.16	0.17	0.17	0.17
			[2.58]	[2.36]	[2.15]	[2.18]	[2.15]	[2.18]
Quintile	Cap	VW	0.20	0.20	0.18	0.18	0.18	0.17
			[2.96]	[3.00]	[2.71]	[2.63]	[2.63]	[2.59]
Decile	NYSE	VW	0.17	0.18	0.19	0.21	0.22	0.24
			[1.69]	[1.79]	[1.88]	[2.11]	[2.20]	[2.37]

Table 3: Conditional sort on size and CNS

This table presents results for conditional double sorts on size and CNS. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on CNS. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high CNS and short stocks with low CNS .Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 196306 to 202406.

Panel A: portfolio average returns and time-series regression results

		CNS Quintiles					CNS Strategies					
		(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Size quintiles	(1)	0.58	0.80	0.82	0.92	0.86	0.28	0.26	0.25	0.21	0.24	0.20
		[2.43]	[3.29]	[3.16]	[3.55]	[3.48]	[3.66]	[3.40]	[3.32]	[2.65]	[3.11]	[2.59]
	(2)	0.77	0.73	0.72	0.91	0.90	0.13	0.13	0.11	0.16	0.14	0.19
		[3.32]	[3.28]	[3.14]	[3.96]	[3.87]	[1.64]	[1.55]	[1.32]	[1.94]	[1.69]	[2.21]
	(3)	0.65	0.74	0.79	0.80	0.89	0.24	0.21	0.19	0.18	0.17	0.17
		[3.23]	[3.64]	[3.78]	[3.87]	[4.11]	[2.94]	[2.49]	[2.32]	[2.23]	[2.09]	[2.07]
	(4)	0.64	0.69	0.70	0.77	0.79	0.16	0.11	0.09	0.07	0.14	0.12
		[3.31]	[3.68]	[3.77]	[3.97]	[3.83]	[1.99]	[1.37]	[1.14]	[0.88]	[1.75]	[1.50]
	(5)	0.43	0.61	0.62	0.49	0.66	0.23	0.26	0.23	0.23	0.21	0.22
		[2.41]	[3.84]	[3.84]	[3.09]	[3.86]	[2.53]	[2.86]	[2.53]	[2.55]	[2.28]	[2.35]

Panel B: Portfolio average number of firms and market capitalization

		CNS Quintiles					CNS Quintiles				
		Average n					Average market capitalization ($\$10^6$)				
		(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)
Size quintiles	(1)	262	263	262	260	259	24	23	21	21	22
	(2)	80	80	80	80	80	43	43	42	42	43
	(3)	62	62	62	61	61	78	78	77	78	78
	(4)	55	55	56	55	55	177	175	183	180	175
	(5)	52	52	52	52	52	1144	1420	1876	1438	1167

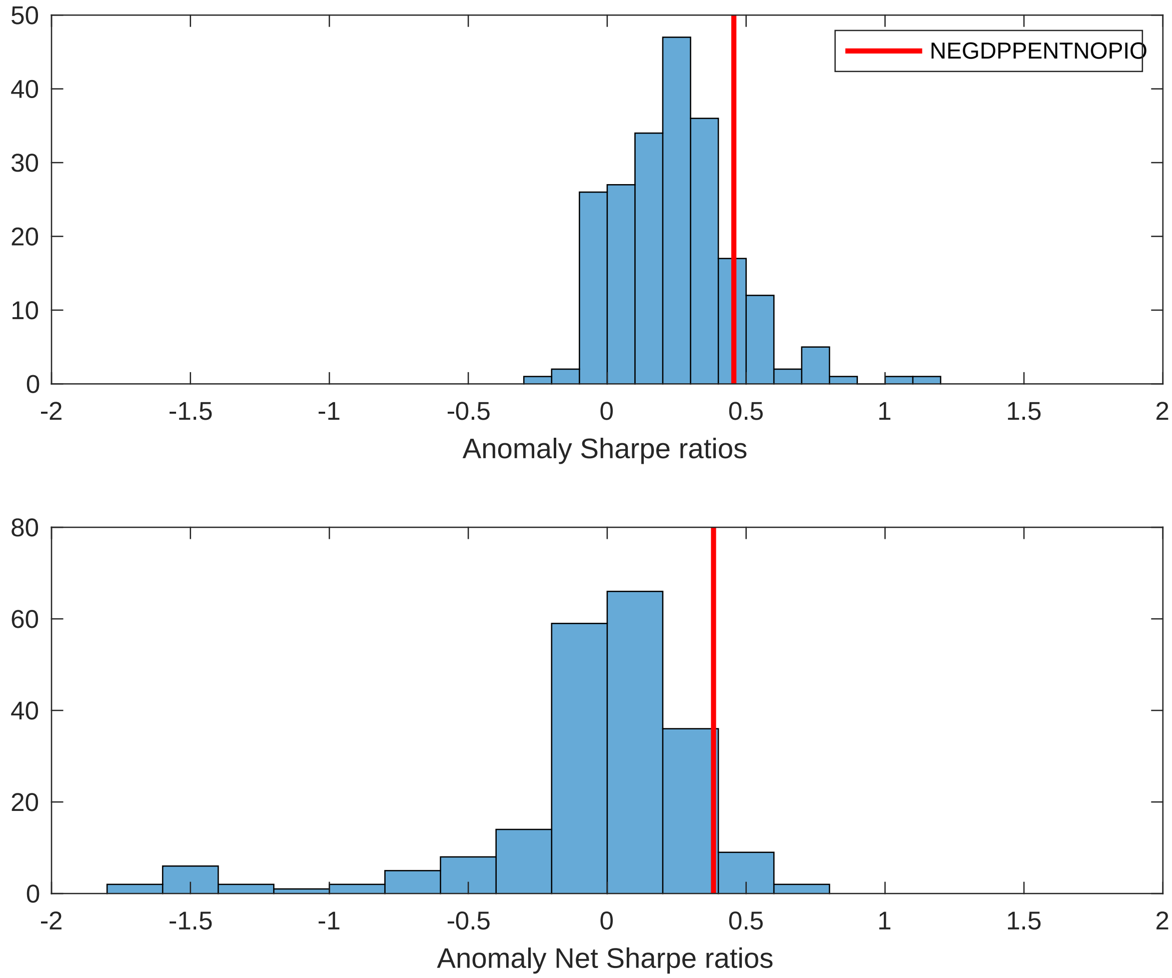


Figure 2: Distribution of Sharpe ratios.
This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the CNS with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

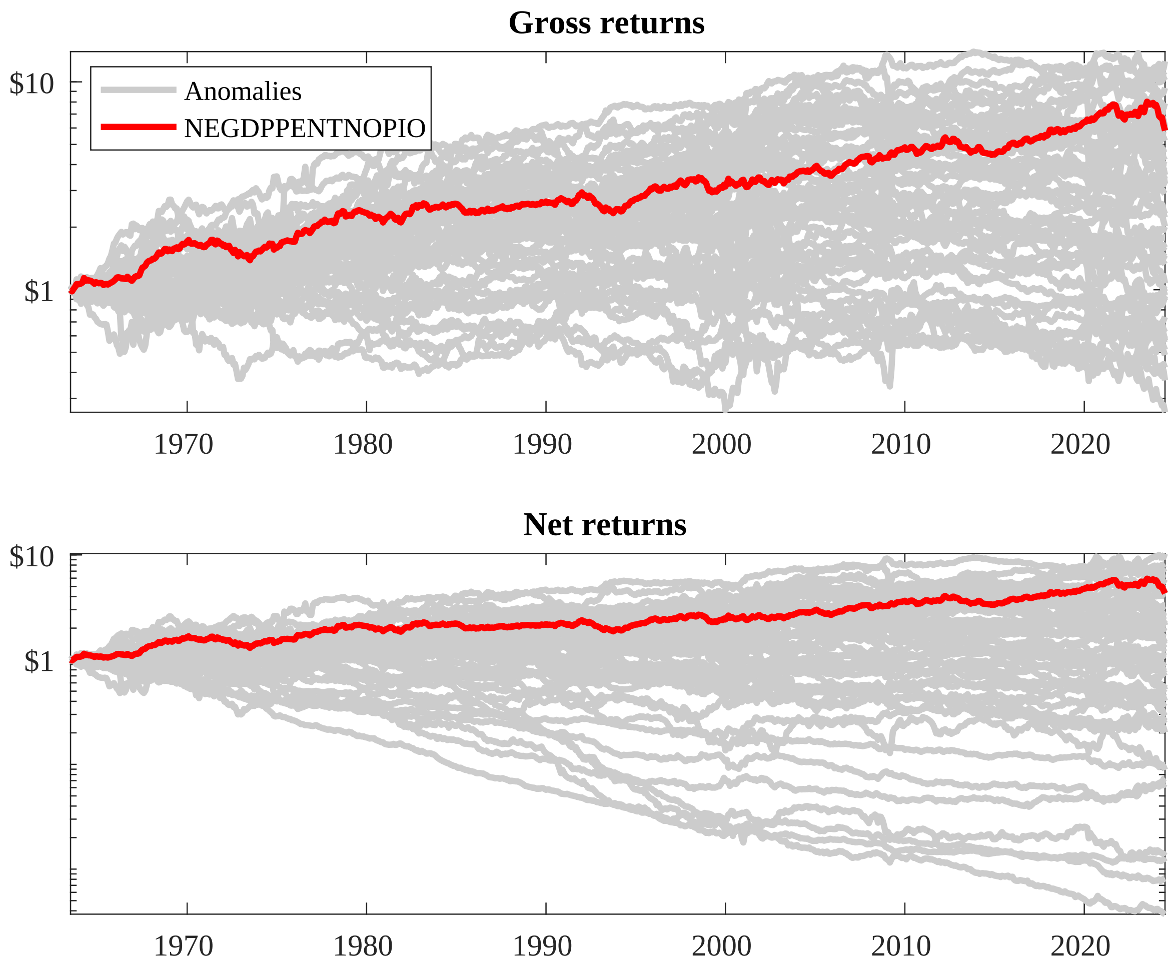
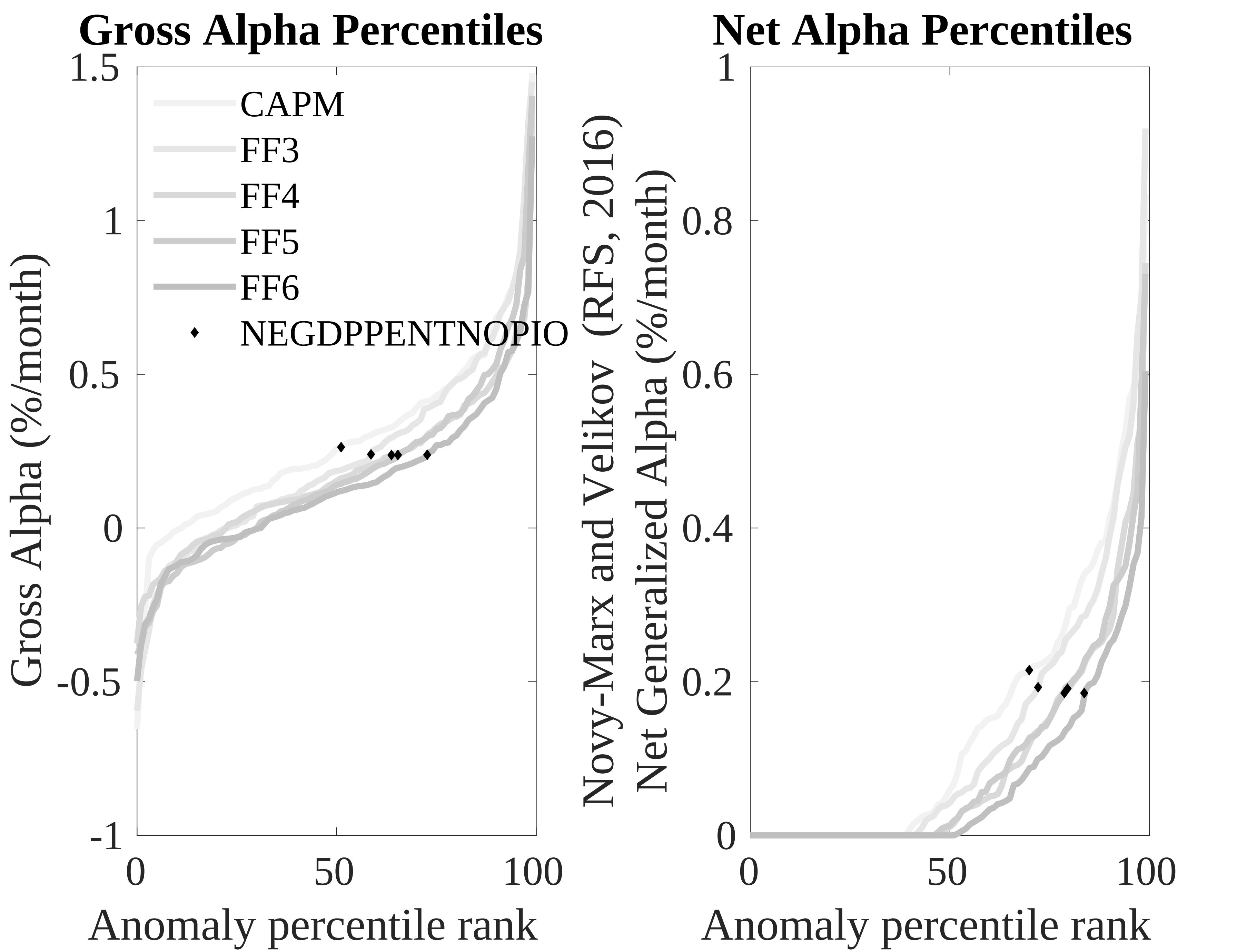
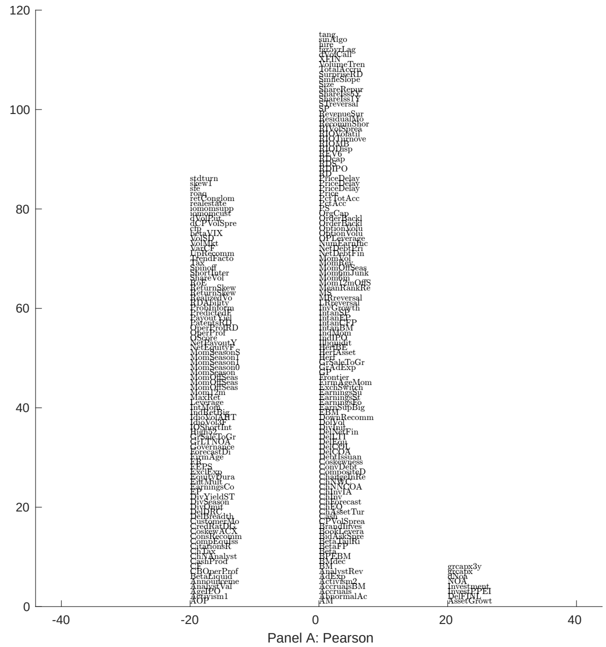


Figure 3: Dollar invested.
This figure plots the growth of a $1 invested in 212 anomaly trading strategies (gray lines), and compares those with the CNS trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strtaegy returns.



Gross Alpha Percentiles
Net Alpha Percentiles
CAPM
FF3
FF4
FF5
FF6
NEGDPPENTNOPIO
Gross Alpha (%/month)
Novy-Marx and Velikov (RFS, 2016)
Net Generalized Alpha (%/month)
Anomaly percentile rank
Anomaly percentile rank
1.5
1
0.5
0
-0.5
-1
0
50
100
1
0.8
0.6
0.4
0.2
0
0
50
100



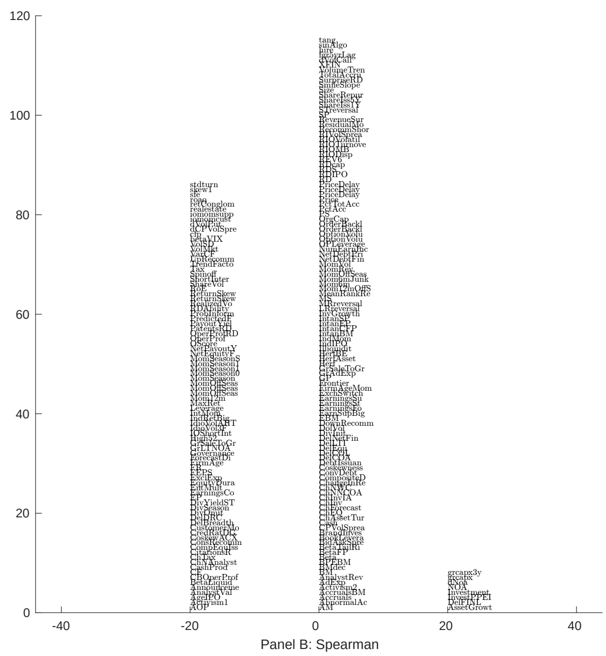


Figure 5: Distribution of correlations.
This figure plots a name histogram of correlations of 209 filtered anomaly signals with CNS. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

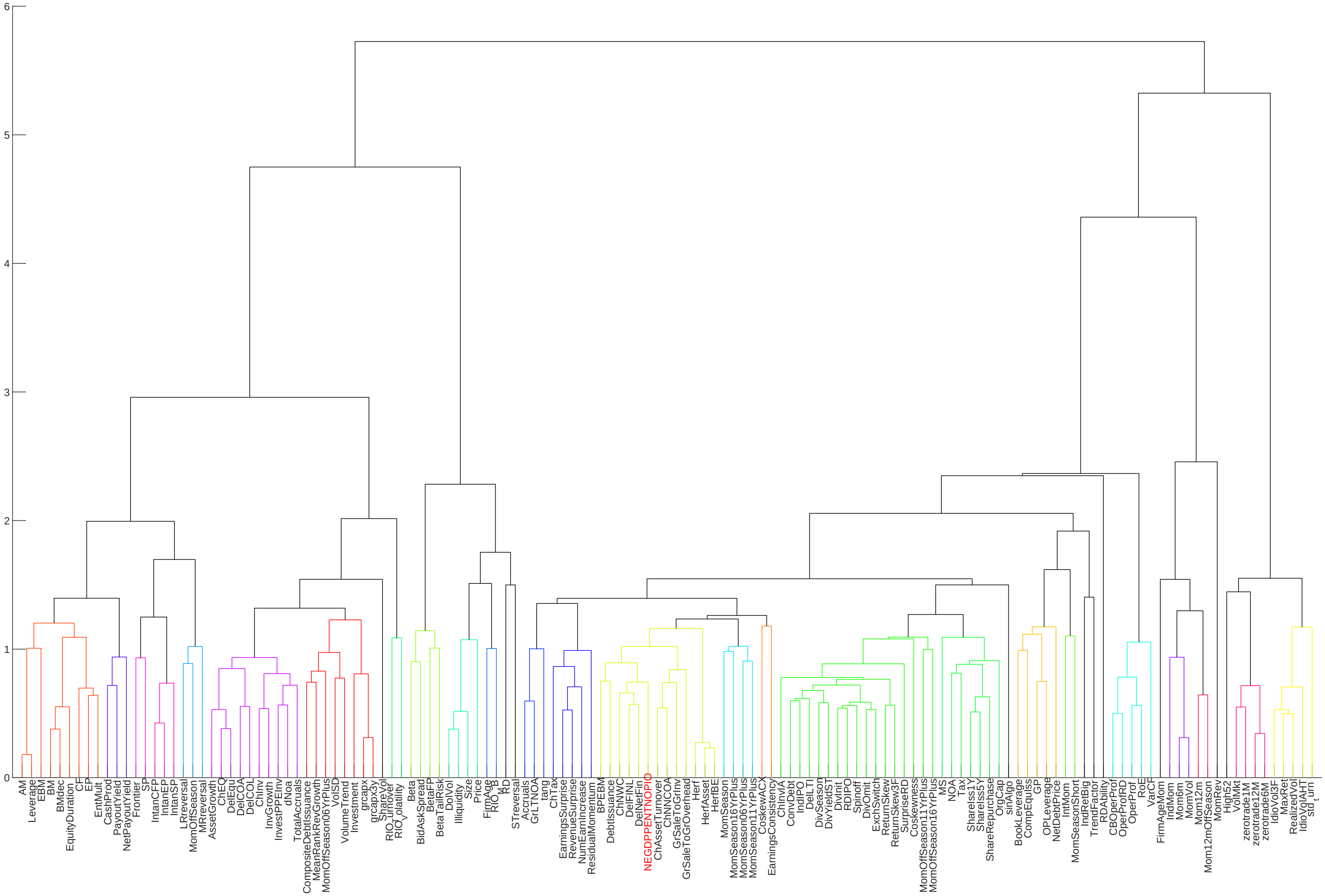


Figure 6: Agglomerative hierarchical cluster plot
This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

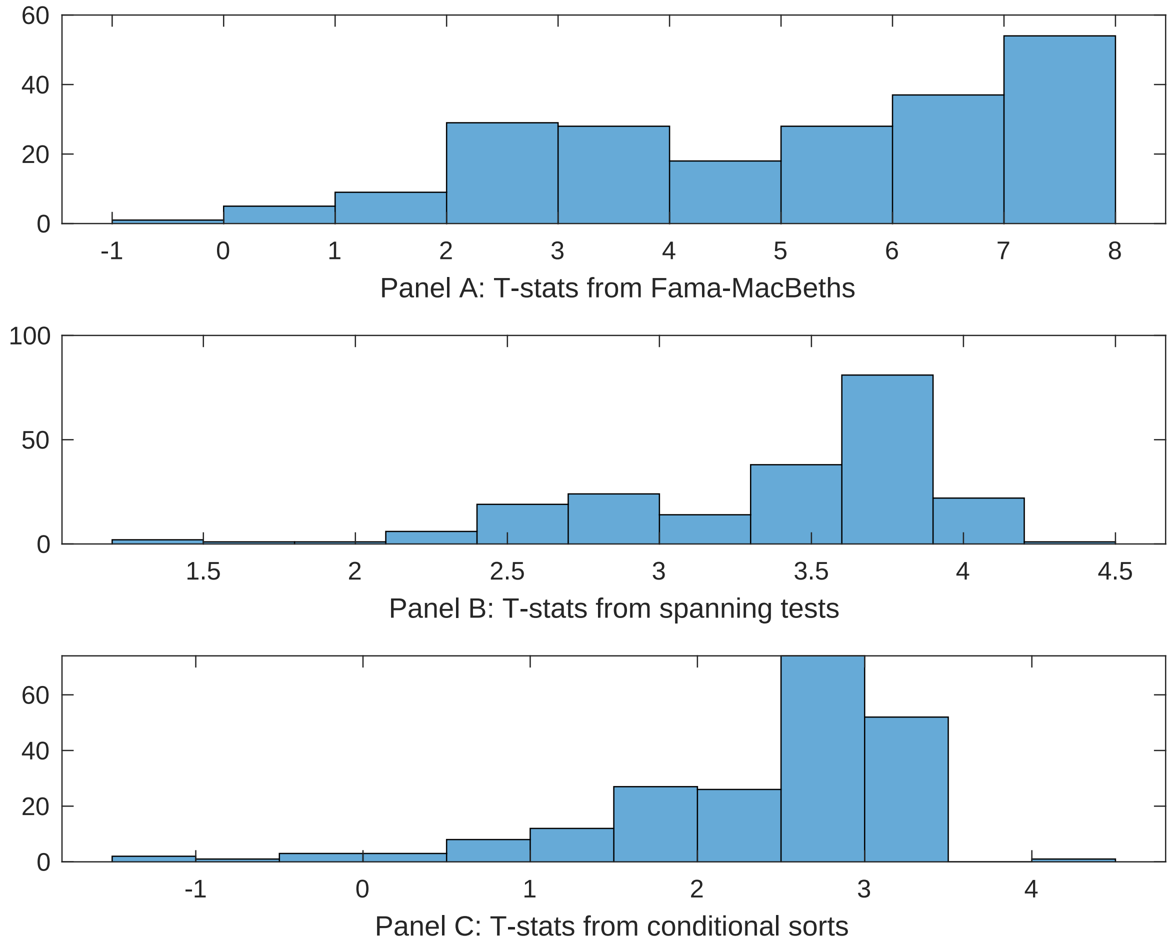


Figure 7: Distribution of t-stats on conditioning strategies
This figure plots histograms of t-statistics for predictability tests of CNS conditioning on each of the 209 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{CNS} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{CNS} CNS_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 209 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{CNS,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 209 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 209 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on CNS. Stocks are finally grouped into five CNS portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted CNS trading strategies conditioned on each of the 209 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on CNS. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{CNS} CNS_{i,t} + \sum_{k=1}^{s} ix\beta_{X_k} X_{i,t}^{k} + \epsilon_{i,t}$. The six most closely related anomalies, X, are change in ppe and inv/assets, change in net operating assets, Change in net financial assets, Net Operating Assets, Change in capex (three years), Inventory Growth. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.13	0.13	0.12	0.17	0.14	0.13	0.15
	[6.08]	[5.82]	[5.59]	[6.99]	[6.61]	[5.68]	[5.48]
CNS	0.63	0.70	0.11	0.94	0.87	0.90	0.20
	[3.66]	[3.71]	[5.93]	[5.37]	[4.38]	[4.20]	[0.89]
Anomaly 1	0.16						0.20
	[8.11]						[0.69]
Anomaly 2		0.13					1.00
		[8.80]					[3.59]
Anomaly 3			0.89				-0.66
			[6.14]				[-2.40]
Anomaly 4				0.80			0.52
				[6.74]			[0.33]
Anomaly 5					0.14		0.83
					[7.60]		[3.91]
Anomaly 6						0.52	0.67
						[9.22]	[0.90]
# months	732	720	732	732	727	732	712
$\bar{R}^2$(%)	0	0	0	0	0	0	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the CNS trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{CNS} = \alpha + \sum_{k=1}^{6} \beta_{X_k} r_t^{X_k} + \sum_{j=1}^{6} \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor. The six most closely related anomalies, X, are change in ppe and inv/assets, change in net operating assets, Change in net financial assets, Net Operating Assets, Change in capex (three years), Inventory Growth. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.24	0.21	0.19	0.19	0.26	0.23	0.20
	[3.21]	[2.72]	[2.42]	[2.44]	[3.33]	[2.99]	[2.66]
Anomaly 1	28.61						23.26
	[8.18]						[5.67]
Anomaly 2		14.10					1.46
		[3.23]					[0.31]
Anomaly 3			19.83				10.62
			[5.36]				[2.84]
Anomaly 4				17.65			10.85
				[5.45]			[3.22]
Anomaly 5					6.61		-5.31
					[1.75]		[-1.37]
Anomaly 6						8.42	-1.77
						[2.69]	[-0.54]
mkt	-0.05	0.38	0.64	-0.33	0.47	0.19	0.17
	[-0.03]	[0.20]	[0.35]	[-0.18]	[0.26]	[0.10]	[0.10]
smb	8.83	10.12	11.12	12.78	8.27	10.53	12.01
	[3.42]	[3.78]	[4.19]	[4.75]	[3.00]	[3.91]	[4.42]
hml	-4.76	-2.72	-1.64	-6.37	-2.74	-2.02	-6.02
	[-1.37]	[-0.76]	[-0.46]	[-1.75]	[-0.76]	[-0.56]	[-1.71]
rmw	-6.49	-6.21	-3.67	-8.16	-6.23	-5.32	-5.49
	[-1.86]	[-1.72]	[-1.02]	[-2.28]	[-1.73]	[-1.46]	[-1.57]
cma	-8.98	2.45	16.43	12.69	8.45	6.07	-0.93
	[-1.57]	[0.40]	[3.14]	[2.44]	[1.51]	[1.03]	[-0.14]
umd	0.11	-0.13	-0.23	-0.47	-0.24	-0.73	-0.04
	[0.06]	[-0.07]	[-0.13]	[-0.26]	[-0.13]	[-0.39]	[-0.02]
# months	732	732	732	732	728	732	728
$\bar{R}^2(\%)$	12	5	7	7	4	5	13

Panel A: Average rank

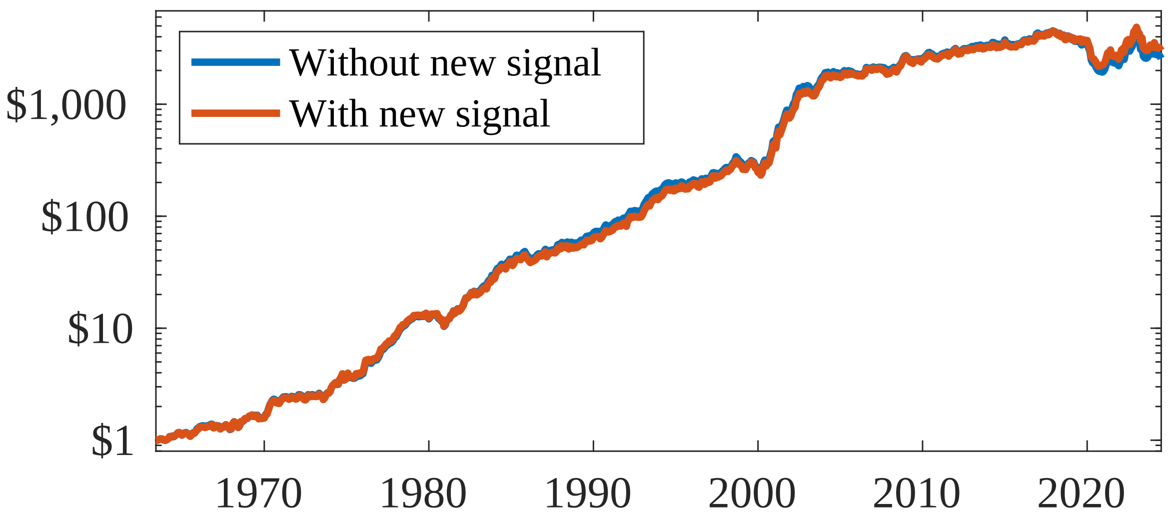


Figure 8: Combination strategy performance
This figure plots the growth of a $1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). In all panels, the blue solid lines indicate combination trading strategies that utilize 154 anomalies. The red solid lines indicate combination trading strategies that utilize the 154 anomalies as well as CNS. Panel A shows results using "Average rank" as the combination method. See Section 7 for details on the combination methods.

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